Advanced Database Management Systems (ADBMS)

Assignment 0 & Assignment 1: Histogram Analysis Report

ADBMS Coursework Submission

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1 Advanced Database Management Systems (ADBMS)

1.1 Assignment 0 & Assignment 1: Histogram Analysis Report

2 Assignment 0

2.1 Setup

- **CPU Model & Specification**: 13th Gen Intel(R) Core(TM) i3-1315U (6 cores, 8 threads)
- **RAM Capacity**: 14 GiB
- **HDD / Storage Capacity**: 456 GiB NVMe SSD
- **Operating System**: Ubuntu 26.04 LTS
- **PostgreSQL Engine Version**: `PostgreSQL 18.6 (Ubuntu 18.6-0ubuntu0.26.04.1) on x86_64-pc-linu compiled by gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0, 64-bit`

2.2 Database

- **Database Name**: Join Order Benchmark (2013 Snapshot)
- **Total Tables**: 21
- **Total Relational Tuples**: 74,190,187

2.2.1 Table Cardinalities & Primary Keys

Table Name	Primary Key Column	Cardinality (`count(*)`)
aka_name	id	901,343
aka_title	id	361,472
cast_info	id	36,244,344
char_name	id	3,140,339
comp_cast_type	id	4
company_name	id	234,997
company_type	id	4
complete_cast	id	135,086
info_type	id	113
keyword	id	134,170
kind_type	id	7
link_type	id	18
movie_companies	id	2,609,129
movie_info	id	14,835,720
movie_info_idx	id	1,380,035
movie_keyword	id	4,523,930
movie_link	id	29,997
name	id	4,167,491
person_info	id	2,963,664
role_type	id	12
title	id	2,528,312

3 Assignment 1: Histogram

3.1 Q1. Histogram Analysis for `id` and `title` Columns of `title` Table

3.1.1 1. Corresponding SQL Query

to see what all colums needed form 'pg_stats'

```
\d pg_stats;
```

to see histogram boundary values and other relevent stored in `pg_stats`:

```
SELECT
    attname,
    null_frac,
    n_distinct,
    most_common_vals,
    most_common_freqs,
    histogram_bounds
FROM pg_stats
WHERE tablename = 'title'
  AND attname IN ('id', 'title');
```

3.1.2 2. Kind of Histogram

psql uses an equi depth histogram , where bucket bounderies are as such so that roughly the same number of tuples in each bucket

3.1.3 3. Number of Buckets

- Column `id`: **100 buckets** (101 bucket boundary values)
- Column `title`: **100 buckets** (101 bucket boundary values)
- extra - to see the defualt number of buckets

```
SHOW default_statistics_target;
```

3.1.4 4. How Textual Attribute `title` is Put into Histogram

textual attribute "title" is put into histogram as:

1. sorting the text strings in lexicographical order like bannana and apple will be apple, bannana after sorting

2. Sampling string values from the table like table title has around 2.5M rows , it will only sample around 30K

3. Encoding initial string prefix bytes into floating point numbers in $[0.0, 1.0]$ range for quantitative interpolation like laptop will be broken to lap with l = 108, a=97, p=112 then by 256 as between 0 and 1 , lap will be around 0.42 , we can then see where a key belongs between two keeys

4. Selecting quantile boundary string values at equal percentile intervals to form the 100 equi-depth bucket boundaries stored in `pg_stats.histogram_bounds`.

3.1.5 5. Exact Bucket Boundaries List (From `pg_stats`)

A. Column `id` Bucket Boundaries (101 values, 100 buckets)

Bucket Index	Boundary Value (`id`)	Bucket Interval
B000 (Start)	71	Start of table data
B001	22629	Bucket 1: [71 - 22629]
B002	47883	Bucket 2: [22629 - 47883]
B003	70297	Bucket 3: [47883 - 70297]
B004	96345	Bucket 4: [70297 - 96345]
B005	122972	Bucket 5: [96345 - 122972]
B006	151571	Bucket 6: [122972 - 151571]
B007	178176	Bucket 7: [151571 - 178176]

B008	202083	Bucket 8: [178176 - 202083]
B009	228264	Bucket 9: [202083 - 228264]
B010	253450	Bucket 10: [228264 - 253450]
B011	277435	Bucket 11: [253450 - 277435]
B012	302244	Bucket 12: [277435 - 302244]
B013	327412	Bucket 13: [302244 - 327412]
B014	351277	Bucket 14: [327412 - 351277]
B015	375269	Bucket 15: [351277 - 375269]
B016	398105	Bucket 16: [375269 - 398105]
B017	419318	Bucket 17: [398105 - 419318]
B018	447243	Bucket 18: [419318 - 447243]
B019	470857	Bucket 19: [447243 - 470857]
B020	497328	Bucket 20: [470857 - 497328]
B021	523098	Bucket 21: [497328 - 523098]
B022	548386	Bucket 22: [523098 - 548386]
B023	573721	Bucket 23: [548386 - 573721]
B024	597600	Bucket 24: [573721 - 597600]
B025	620953	Bucket 25: [597600 - 620953]
B026	644334	Bucket 26: [620953 - 644334]
B027	669359	Bucket 27: [644334 - 669359]
B028	694359	Bucket 28: [669359 - 694359]
B029	719940	Bucket 29: [694359 - 719940]
B030	746531	Bucket 30: [719940 - 746531]
B031	772412	Bucket 31: [746531 - 772412]
B032	799868	Bucket 32: [772412 - 799868]
B033	822650	Bucket 33: [799868 - 822650]
B034	846897	Bucket 34: [822650 - 846897]
B035	872174	Bucket 35: [846897 - 872174]
B036	897463	Bucket 36: [872174 - 897463]
B037	923221	Bucket 37: [897463 - 923221]
B038	945908	Bucket 38: [923221 - 945908]
B039	973357	Bucket 39: [945908 - 973357]
B040	1000734	Bucket 40: [973357 - 1000734]
B041	1023461	Bucket 41: [1000734 - 1023461]
B042	1048673	Bucket 42: [1023461 - 1048673]
B043	1071915	Bucket 43: [1048673 - 1071915]
B044	1100137	Bucket 44: [1071915 - 1100137]
B045	1124383	Bucket 45: [1100137 - 1124383]
B046	1149120	Bucket 46: [1124383 - 1149120]
B047	1175291	Bucket 47: [1149120 - 1175291]
B048	1201852	Bucket 48: [1175291 - 1201852]
B049	1228957	Bucket 49: [1201852 - 1228957]
B050	1257141	Bucket 50: [1228957 - 1257141]
B051	1284123	Bucket 51: [1257141 - 1284123]
B052	1308942	Bucket 52: [1284123 - 1308942]
B053	1335382	Bucket 53: [1308942 - 1335382]
B054	1361256	Bucket 54: [1335382 - 1361256]
B055	1387631	Bucket 55: [1361256 - 1387631]
B056	1411606	Bucket 56: [1387631 - 1411606]
B057	1436629	Bucket 57: [1411606 - 1436629]
B058	1461485	Bucket 58: [1436629 - 1461485]
B059	1487317	Bucket 59: [1461485 - 1487317]
B060	1510970	Bucket 60: [1487317 - 1510970]
B061	1533534	Bucket 61: [1510970 - 1533534]
B062	1560080	Bucket 62: [1533534 - 1560080]
B063	1586138	Bucket 63: [1560080 - 1586138]
B064	1612928	Bucket 64: [1586138 - 1612928]
B065	1638329	Bucket 65: [1612928 - 1638329]
B066	1661883	Bucket 66: [1638329 - 1661883]
B067	1686067	Bucket 67: [1661883 - 1686067]

B068	1711184	Bucket 68: [1686067 - 1711184]
B069	1735833	Bucket 69: [1711184 - 1735833]
B070	1763217	Bucket 70: [1735833 - 1763217]
B071	1788453	Bucket 71: [1763217 - 1788453]
B072	1814568	Bucket 72: [1788453 - 1814568]
B073	1836400	Bucket 73: [1814568 - 1836400]
B074	1861486	Bucket 74: [1836400 - 1861486]
B075	1887550	Bucket 75: [1861486 - 1887550]
B076	1914087	Bucket 76: [1887550 - 1914087]
B077	1938329	Bucket 77: [1914087 - 1938329]
B078	1962799	Bucket 78: [1938329 - 1962799]
B079	1988458	Bucket 79: [1962799 - 1988458]
B080	2014885	Bucket 80: [1988458 - 2014885]
B081	2038757	Bucket 81: [2014885 - 2038757]
B082	2064538	Bucket 82: [2038757 - 2064538]
B083	2089516	Bucket 83: [2064538 - 2089516]
B084	2115434	Bucket 84: [2089516 - 2115434]
B085	2140641	Bucket 85: [2115434 - 2140641]
B086	2166812	Bucket 86: [2140641 - 2166812]
B087	2190040	Bucket 87: [2166812 - 2190040]
B088	2218367	Bucket 88: [2190040 - 2218367]
B089	2246129	Bucket 89: [2218367 - 2246129]
B090	2270048	Bucket 90: [2246129 - 2270048]
B091	2295684	Bucket 91: [2270048 - 2295684]
B092	2322768	Bucket 92: [2295684 - 2322768]
B093	2345812	Bucket 93: [2322768 - 2345812]
B094	2373919	Bucket 94: [2345812 - 2373919]
B095	2397988	Bucket 95: [2373919 - 2397988]
B096	2425418	Bucket 96: [2397988 - 2425418]
B097	2449900	Bucket 97: [2425418 - 2449900]
B098	2475221	Bucket 98: [2449900 - 2475221]
B099	2502193	Bucket 99: [2475221 - 2502193]
B100	2528277	Bucket 100: [2502193 - 2528277]

B. Column `title` Bucket Boundaries (101 values, 100 buckets)

Bucket Index	Boundary Value (`title`)	Lexicographical Interval
B000 (Start)	`\$25 Million Dollar Hoax`	Start of text data
B001	`(#1.126)`	Bucket 1: [`\$25 Million Dollar Hoax` - `(#1.126)`]
B002	`(#1.200)`	Bucket 2: [`(#1.126)` - `(#1.200)`]
B003	`(#13.49)`	Bucket 3: [`(#1.200)` - `(#13.49)`]
B004	`(#1.5527)`	Bucket 4: [`(#13.49)` - `(#1.5527)`]
B005	`(#1.765)`	Bucket 5: [`(#1.5527)` - `(#1.765)`]
B006	`(1961-05-01)`	Bucket 6: [`(#1.765)` - `(1961-05-01)`]
B007	`(1978-12-09)`	Bucket 7: [`(1961-05-01)` - `(1978-12-09)`]
B008	`(1993-01-04)`	Bucket 8: [`(1978-12-09)` - `(1993-01-04)`]
B009	`(1997-01-16)`	Bucket 9: [`(1993-01-04)` - `(1997-01-16)`]
B010	`(1999-07-05)`	Bucket 10: [`(1997-01-16)` - `(1999-07-05)`]
B011	`(2001-05-05)`	Bucket 11: [`(1999-07-05)` - `(2001-05-05)`]
B012	`(2003-03-11)`	Bucket 12: [`(2001-05-05)` - `(2003-03-11)`]
B013	`(2004-09-18)`	Bucket 13: [`(2003-03-11)` - `(2004-09-18)`]
B014	`(2005-12-07)`	Bucket 14: [`(2004-09-18)` - `(2005-12-07)`]
B015	`(2007-05-09)`	Bucket 15: [`(2005-12-07)` - `(2007-05-09)`]
B016	`(2008-10-29)`	Bucket 16: [`(2007-05-09)` - `(2008-10-29)`]
B017	`(2010-06-15)`	Bucket 17: [`(2008-10-29)` - `(2010-06-15)`]
B018	`(2011-09-20)`	Bucket 18: [`(2010-06-15)` - `(2011-09-20)`]
B019	`(2012-08-16)`	Bucket 19: [`(2011-09-20)` - `(2012-08-16)`]
B020	`(#2.112)`	Bucket 20: [`(2012-08-16)` - `(#2.112)`]

B021	(#3.120)	Bucket 21: [(#2.112) - (#3.120)]
B022	(#4.84)	Bucket 22: [(#3.120) - (#4.84)]
B023	(#7.13)	Bucket 23: [(#4.84) - (#7.13)]
B024	Abschiede	Bucket 24: [(#7.13) - Abschiede]
B025	After All I've Said!	Bucket 25: [Abschiede - After All I've Said!]
B026	All for Love	Bucket 26: [After All I've Said! - All for Love]
B027	And the Wind Cries... Wendigo	Bucket 27: [All for Love - And the Wind Cries... Wendigo]
B028	Arena total 12 - Volle Spermabacken	Bucket 28: [And the Wind Cries... Wendigo - Arena total 12 - Volle Spermabacken]
B029	Auf den Hund gekommen	Bucket 29: [Arena total 12 - Volle Spermabacken - Auf den Hund gekommen]
B030	Bandaged	Bucket 30: [Auf den Hund gekommen - Bandaged]
B031	Best Neighbor	Bucket 31: [Bandaged - Best Neighbor]
B032	Blood Bank	Bucket 32: [Best Neighbor - Blood Bank]
B033	Brudekronen	Bucket 33: [Blood Bank - Brudekronen]
B034	Care Bears: The Giving Festival Movie	Bucket 34: [Brudekronen - Care Bears: The Giving Festival Movie]
B035	Chief Guard Saldeath's Trap! Level 3: Starvation Hell!	Bucket 35: [Care Bears: The Giving Festival Movie - Chief Guard Saldeath's Trap! Level 3: Starvation Hell!]
B036	Come and Get It	Bucket 36: [Chief Guard Saldeath's Trap! Level 3: Starvation Hell! - Come and Get It]
B037	Crni fond	Bucket 37: [Come and Get It - Crni fond]
B038	Das ist mein Tagwerk	Bucket 38: [Crni fond - Das ist mein Tagwerk]
B039	Delta Queen	Bucket 39: [Das ist mein Tagwerk - Delta Queen]
B040	Deux filles en cavale	Bucket 40: [Delta Queen - Deux filles en cavale]
B041	Dishin on Dishes	Bucket 41: [Deux filles en cavale - Dishin on Dishes]
B042	Drift	Bucket 42: [Dishin on Dishes - Drift]
B043	El Aborto	Bucket 43: [Drift - El Aborto]
B044	Ensayo de una noche de bodas	Bucket 44: [El Aborto - Ensayo de una noche de bodas]
B045	Fake Orgasm	Bucket 45: [Ensayo de una noche de bodas - Fake Orgasm]
B046	First Place	Bucket 46: [Fake Orgasm - First Place]
B047	Freeride World Tour	Bucket 47: [First Place - Freeride World Tour]
B048	Gerhard Haderer - Der Kult-Karikaturist	Bucket 48: [Freeride World Tour - Gerhard Haderer - Der Kult-Karikaturist]
B049	Grad Nite	Bucket 49: [Gerhard Haderer - Der Kult-Karikaturist - Grad Nite]
B050	Hanging On	Bucket 50: [Grad Nite - Hanging On]
B051	Herz über Kopf	Bucket 51: [Hanging On - Herz über Kopf]
B052	Horny College Schoolgirls	Bucket 52: [Herz über Kopf - Horny College Schoolgirls]
B053	I Hope I Am Crazy	Bucket 53: [Horny College Schoolgirls - I Hope I Am Crazy]
B054	Instinto fatal	Bucket 54: [I Hope I Am Crazy - Instinto fatal]

B055	Jai Bajrang	Bucket 55: [Instinto fatal - Jai Bajrang]
B056	Judgment Day	Bucket 56: [Jai Bajrang - Judgment Day]
B057	Kenny Chesney: Road Case - The Movie	Bucket 57: [Judgment Day - Kenny Chesney: Road Case - The Movie]
B058	Kreuzfeuer	Bucket 58: [Kenny Chesney: Road Case - The Movie - Kreuzfeuer]
B059	La mitad	Bucket 59: [Kreuzfeuer - La mitad]
B060	LeBron vs Kobe	Bucket 60: [La mitad - LeBron vs Kobe]
B061	Le trait d'union	Bucket 61: [LeBron vs Kobe - Le trait d'union]
B062	Long Time Legends	Bucket 62: [Le trait d'union - Long Time Legends]
B063	Macula	Bucket 63: [Long Time Legends - Macula]
B064	Máscaras mojadas	Bucket 64: [Macula - Máscaras mojadas]
B065	Midnight Eyes	Bucket 65: [Máscaras mojadas - Midnight Eyes]
B066	Monsters Inside Me	Bucket 66: [Midnight Eyes - Monsters Inside Me]
B067	My Name Is Manolete	Bucket 67: [Monsters Inside Me - My Name Is Manolete]
B068	New York Fight Clubs	Bucket 68: [My Name Is Manolete - New York Fight Clubs]
B069	Nun va Goldoon	Bucket 69: [New York Fight Clubs - Nun va Goldoon]
B070	Only Schmucks Pay Income Tax	Bucket 70: [Nun va Goldoon - Only Schmucks Pay Income Tax]
B071	Paidali Padleli Phule	Bucket 71: [Only Schmucks Pay Income Tax - Paidali Padleli Phule]
B072	Peter Townend	Bucket 72: [Paidali Padleli Phule - Peter Townend]
B073	Pop rendez-vous	Bucket 73: [Peter Townend - Pop rendez-vous]
B074	Punto y raya	Bucket 74: [Pop rendez-vous - Punto y raya]
B075	Red Boot Diaries	Bucket 75: [Punto y raya - Red Boot Diaries]
B076	Rocket Dog	Bucket 76: [Red Boot Diaries - Rocket Dog]
B077	Sammy Hagar	Bucket 77: [Rocket Dog - Sammy Hagar]
B078	Secret Life of Cats	Bucket 78: [Sammy Hagar - Secret Life of Cats]
B079	Shihe no meikyu	Bucket 79: [Secret Life of Cats - Shihe no meikyu]
B080	Sluby i rozstania	Bucket 80: [Shihe no meikyu - Sluby i rozstania]
B081	Splintered	Bucket 81: [Sluby i rozstania - Splintered]
B082	Sucre noir	Bucket 82: [Splintered - Sucre noir]
B083	Take Down	Bucket 83: [Sucre noir - Take Down]
B084	The \$1.98 Beauty Show \vert Bucket 84: [Take Down - The \$1.98 Beauty Show]	
B085	The Brewing of Coffee Date	Bucket 85: [The \$1.98 Beauty Show - The Brewing of Coffee Date]
B086	The Diamond Smugglers!	Bucket 86: [The Brewing of Coffee Date - The Diamond Smugglers!]
B087	The Gray Seasons	Bucket 87: [The Diamond Smugglers! - The Gray Seasons]
B088	The Last Superhero	Bucket 88: [The Gray Seasons - The Last Superhero]

B089	The Murderer	Bucket 89: [The Last Superhero - The Murderer]
B090	The Real Deal	Bucket 90: [The Murderer - The Real Deal]
B091	The Student Who Never Was	Bucket 91: [The Real Deal - The Student Who Never Was]
B092	The World Through the Eyes of Children	Bucket 92: [The Student Who Never Was - The World Through the Eyes of Children]
B093	To Mars...	Bucket 93: [The World Through the Eyes of Children - To Mars...]
B094	Troubled Waters	Bucket 94: [To Mars... - Troubled Waters]
B095	Understudies	Bucket 95: [Troubled Waters - Understudies]
B096	Venecia	Bucket 96: [Understudies - Venecia]
B097	Walt and Lavinia	Bucket 97: [Venecia - Walt and Lavinia]
B098	While Shepherds Watched	Bucket 98: [Walt and Lavinia - While Shepherds Watched]
B099	Xin leng xue shi san ying	Bucket 99: [While Shepherds Watched - Xin leng xue shi san ying]
B100	ZZ Top: Sleeping Bag	Bucket 100: [Xin leng xue shi san ying - ZZ Top: Sleeping Bag]

3.1.6 6. Histogram Visualization

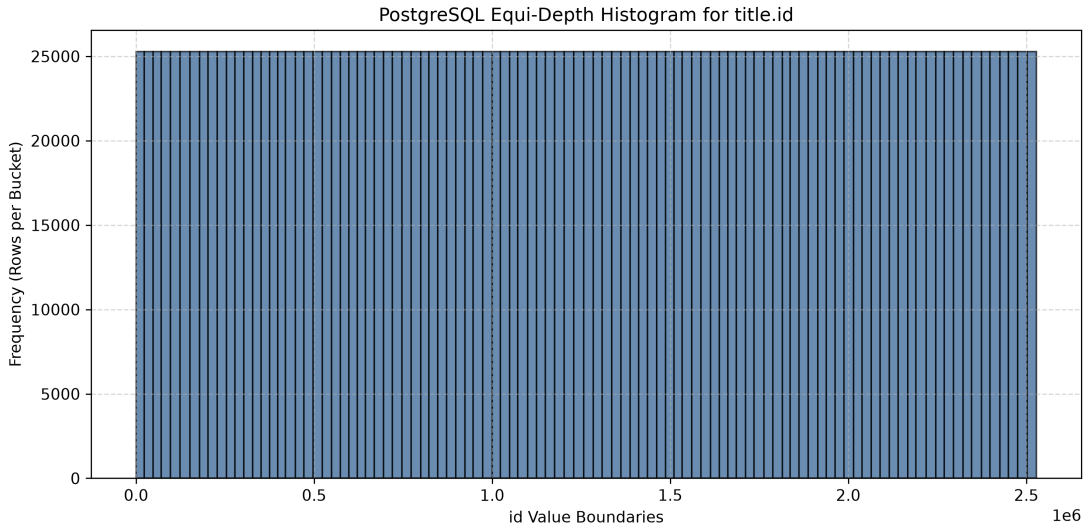


Figure 1: Histogram for title.id

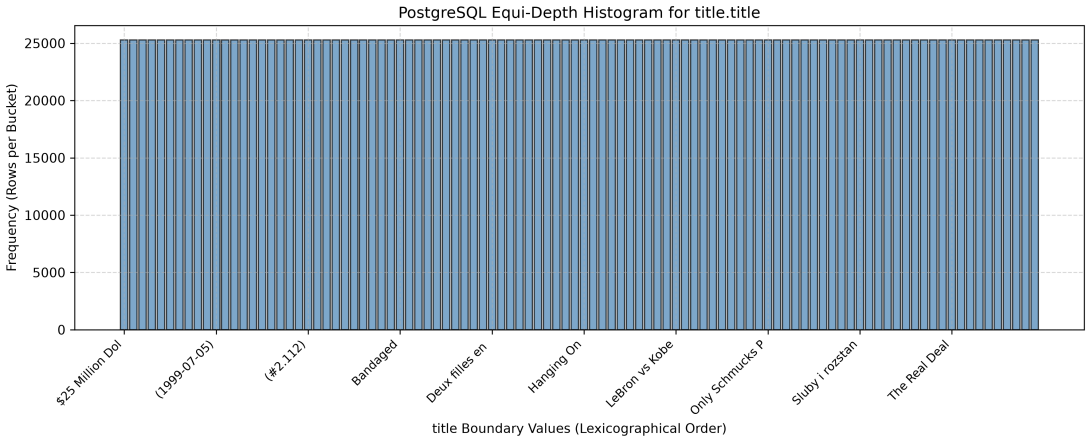


Figure 2: Histogram for title.title

3.1.7 7. Max Selectivity Error

- **Max Selectivity Error**: Since `default_statistics_target` is 100, there are 100 buckets. Therefore, the maximum selectivity error for a range query is bounded by:

$$\text{Max Error} \leq \frac{1}{B} = \frac{1}{100} = 0.01 \quad (1.0\%)$$

For evaluated queries(q9), the observed maximum selectivity error is **0.8592%** (0.008592), which strictly satisfies Max Error $\leq 1.0\%$.

3.1.8 8: Selectivity Estimation for Text Columns

To estimate selectivity for a range query on a text column like title, psql first looks at the histogram-bounds array in pgstats to figure out which bucket contains the query string, once it finds the target bucket, it counts all the full buckets that come before it. since it uses an equidepth histogram, each full bucket represents a fixed fraction of the total rows. For the specific bucket where the string actually falls, psql assumes the data is uniformly distributed and uses linear interpolation. It converts the first few bytes of the strings into floating point numbers between 0 and 1. This allows it to calculate the mathematical fraction of exactly where the query string sits between the buckets minimum and maximum boundaries. finally it adds this fractional piece to the sum of the preceding full buckets to get the final overall selectivity estimate

3.1.9 9: Estimated vs. Actual Selectivities

To calculate estimated and actual selectivities of the queries and moreover there accuracy we compare estimated rows (from `EXPLAIN`) against actual rows (from `EXPLAIN ANALYZE`).

Total row count (N) for the `title` table is 2,528,312.

Query 1: `title < 'Race'`

```
EXPLAIN ANALYZE SELECT * FROM title WHERE title.title < 'Race';
```

- **Actual Selectivity:** The query physically returned 1,918,141 matching rows: $S_{\text{act}} = \frac{1918141}{2528312} \approx 0.7587$ (75.87%).

```
EXPLAIN SELECT * FROM title WHERE title.title < 'Race';
```

- **Estimated Selectivity:** The planner estimated 1,939,865 rows: $S_{\text{est}} = \frac{1939865}{2528312} \approx 0.7673$ (76.73%).
- **Observation:** The absolute error is roughly 0.86%.

Query 2: `id < 350000`

```
EXPLAIN ANALYZE SELECT * FROM title WHERE title.id < '350000';
```

- **Actual Selectivity:** The query returned 349,999 matching rows: $S_{\text{act}} = \frac{349999}{2528312} \approx 0.1384$ (13.84%).

```
EXPLAIN SELECT * FROM title WHERE title.id < '350000';
```

- **Estimated Selectivity:** The planner estimated 352,649 rows: $S_{\text{est}} = \frac{352649}{2528312} \approx 0.1395$ (13.95%).
- **Observation:** The absolute error here is about 0.10%.

3.2 Q2. Optimal Serial Histogram Construction ($B = 10$ Buckets)

3.2.1 1. Histogram Visualizations for Exact $N = 3000$ Samples (Slide-9 Upright Vertical Block Plots)

As defined in lecture Slide 9, a Serial Histogram sorts domain values by their ordered frequencies $(f(v_{(1)}) \leq f(v_{(2)}) \leq \cdots \leq f(v_{(m)}))$ and partitions the frequency space to minimize frequency variance $\sum n_i V_i$. The upright vertical block plots below feature Frequencies on the X-axis (ordered ascending) and Value Set / Stacked Domain Elements on the Y-axis:

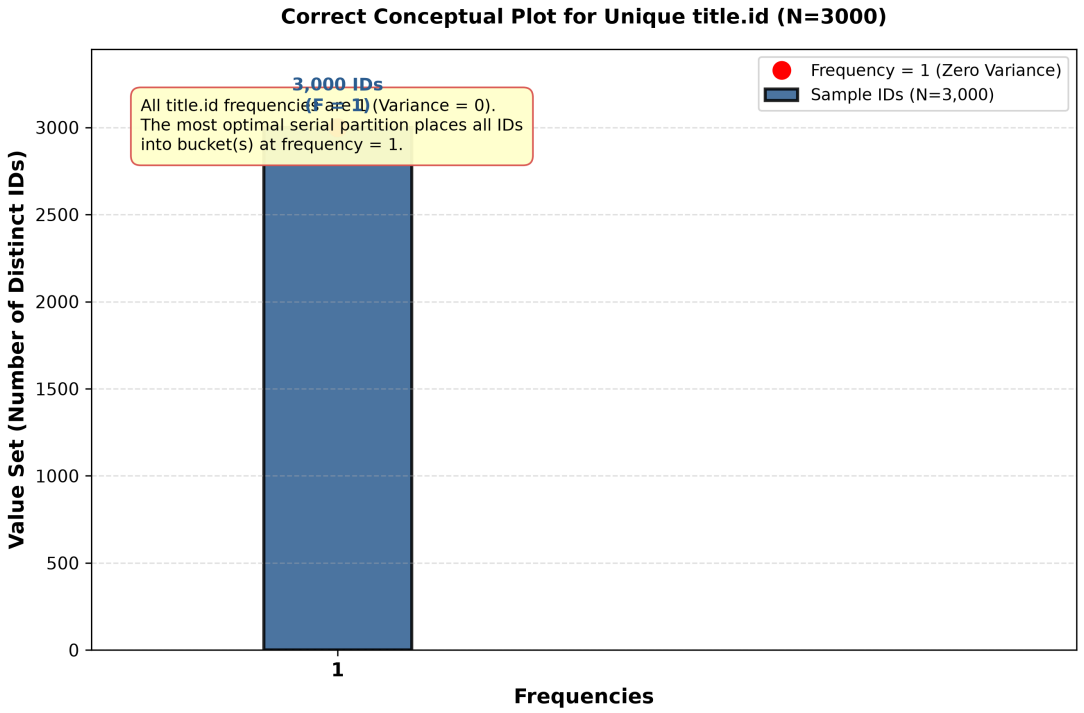


Figure 3: Correct Conceptual Plot for Unique title.id (N=3000)

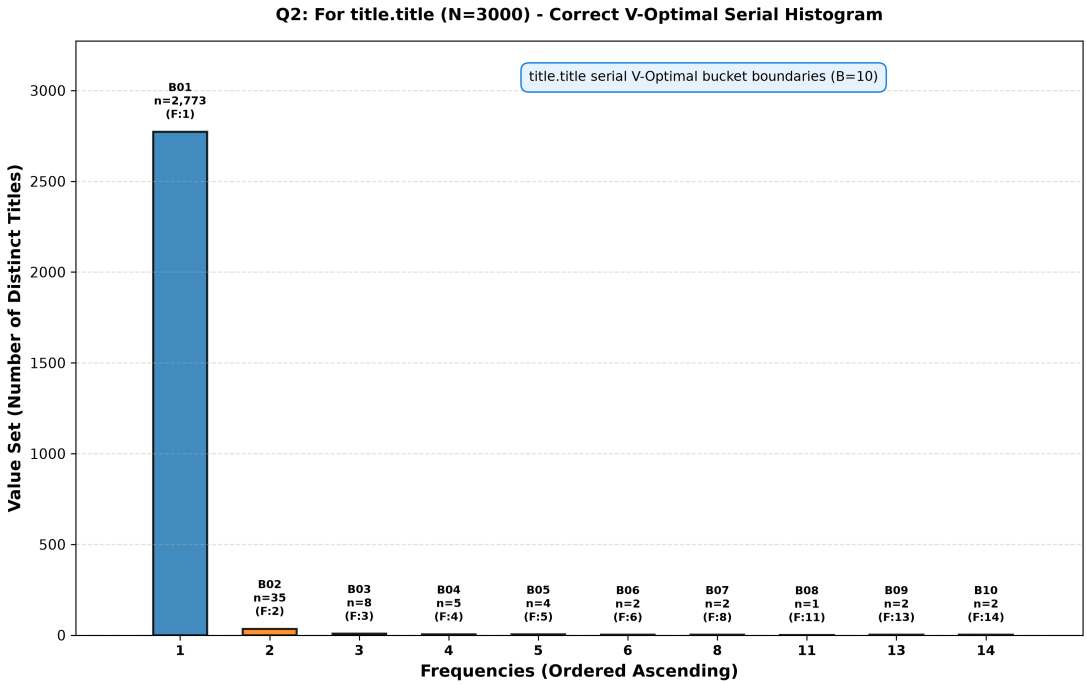


Figure 4: Q2: For title.title (N=3000) - Correct V-Optimal Serial Histogram

3.2.2 2. Execution Build Time (Average over 3 Runs)

- **Average Build Time for $N = 3000$: 0.0018 seconds **(`id` column)** / 0.0048 seconds** (`title` column)

3.2.3 3. Frequency Bucket Boundaries ($N = 3000$, $B = 10$)

The optimal serial histogram algorithm partitions the exact 3,000 samples into dynamic frequency buckets ($B = 10$). The exact bucket value ranges, distinct domain value counts (n_i), and frequency distributions for both columns are listed below:

A. Column `id` Serial Frequency Bucket Boundaries ($N = 3000$, $B = 10$)

Bucket #	Value Range (`id`)	Distinct Values (n_i)	Frequency Range	Average Frequency (arf_i)
Bucket 01	[1,234 - 2,528,274]	3000	1	1.00

B. Column `title` Serial Frequency Bucket Boundaries ($N = 3000$, $B = 10$)

Bucket #	Value Range (`title`)	Distinct Values (n_i)	Frequency Range	Average Frequency (arf_i)
Bucket 01	["\$#!+ Happens" ... "Ônna no kaôyaku"]	2773	1	1.00
Bucket 02	["(#1.132)" ... "The Box"]	35	2	2.00
Bucket 03	["(#1.118)" ... "(#4.1)"]	8	3	3.00
Bucket 04	["(#1.11)" ... "(#3.1)"]	5	4	4.00
Bucket 05	["(#1.10)" ... "Pilot"]	4	5	5.00

Bucket 06	["(#1.7)" ... "(#1.9)"]	2	6	6.00
Bucket 07	["(#1.6)" ... "(#1.8)"]	2	8	8.00
Bucket 08	["(#1.5)" ... "(#1.5)"]	1	11	11.00
Bucket 09	["(#1.1)" ... "(#1.2)"]	2	13	13.00
Bucket 10	["(#1.3)" ... "(#1.4)"]	2	14	14.00

3.2.4 4. 3-Run Selectivity Error Calculation & Average ($N = 3000$)

As required to account for sample randomness, the selectivity error evaluation was executed across **3 independent runs** (with exact sample size $N = 3000$). All three run errors are listed below along with their computed average:

A. 3-Run Max Selectivity Error Summary ($N = 3000$)

Column	Run 1 Max Error	Run 2 Max Error	Run 3 Max Error	Average Max Selectivity Error
id	0.022444	0.009556	0.009889	0.013963 (1.3963%)
title	0.612556	0.609000	0.616667	0.612741 (61.2741%)

B. Quantile-by-Quantile Error Evaluation Table for Run 1 ($N = 3000, B = 10$)

Index (k)	Query Threshold (q_k)	Actual Count (A)	Actual Selectivity (S_{act})	Estimated Count (E)	Estimated Selectivity (S_{est})	Absolute Error $\lvert S_{\text{est}} - S_{\text{act}} \rvert$
q_0	1,234.0	1	0.000333	0.00	0.000000	0.000333
q_1	282,016.2	352	0.117333	333.33	0.111111	0.006222
q_2	562,798.4	673	0.224333	666.67	0.222222	0.002111
q_3	843,580.7	966	0.322000	1000.00	0.333333	0.011333
q_4	1,124,362.9	1307	0.435667	1333.33	0.444444	0.008778
q_5	1,405,145.1	1613	0.537667	1666.67	0.555556	0.017889
q_6	1,685,927.3	1946	0.648667	2000.00	0.666667	0.018000
q_7	1,966,709.6	2266	0.755333	2333.33	0.777778	**0.022444**
q_8	2,247,491.8	2623	0.874333	2666.67	0.888889	0.014556
q_9	2,528,274.0	3000	1.000000	3000.00	1.000000	0.000000

3.2.5 5. Histograms Across Exact Sample Sizes ($N = 1000, 3000, 5000$)

Below are the optimal serial histogram visualizations ($B = 10$) for exact sample sizes $N = 1000$, $N = 3000$, and $N = 5000$:N=5000$:

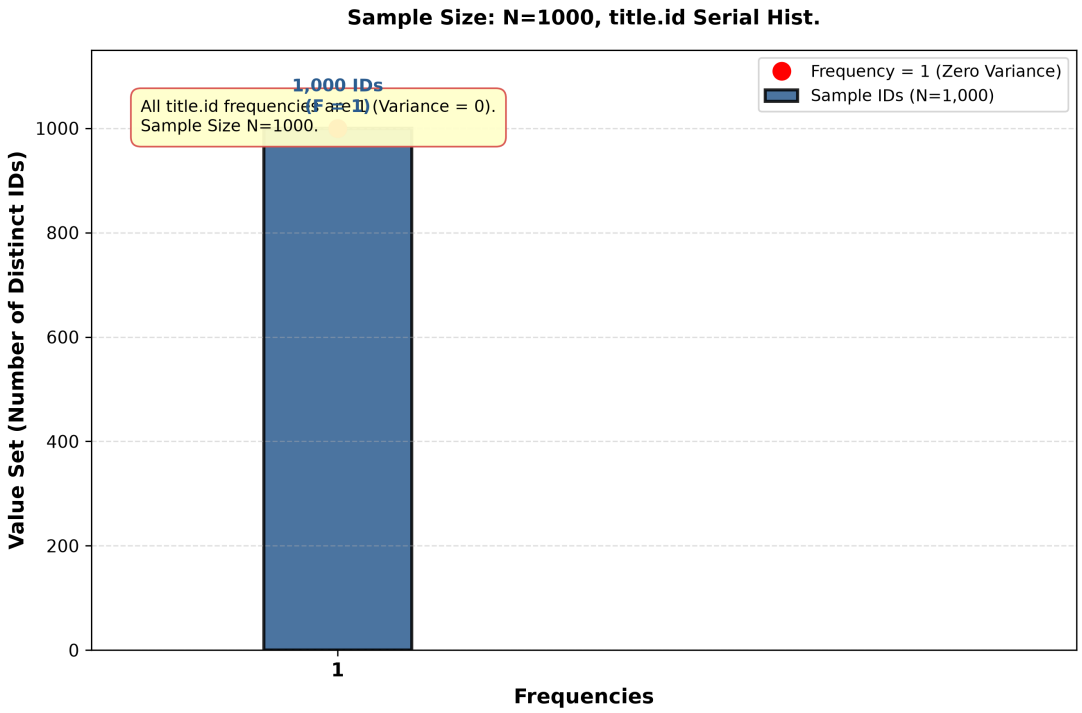


Figure 5: Sample Size: N=1000, title.id Serial Hist.

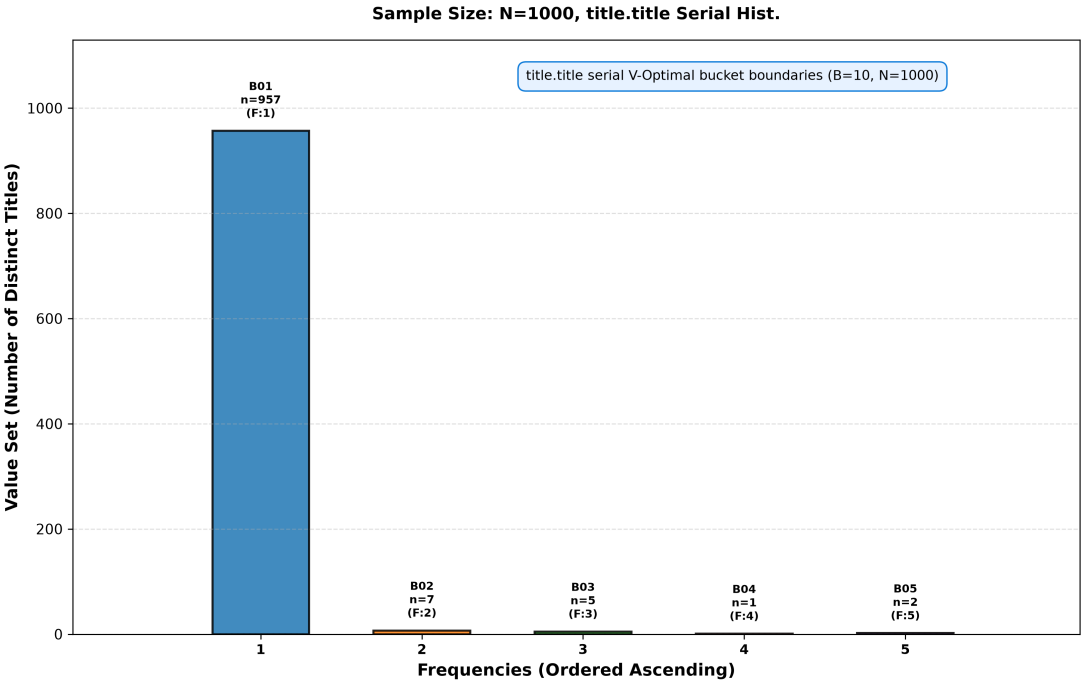


Figure 6: Sample Size: N=1000, title.title Serial Hist.

A. Sample Size $N = 1000$ Histograms

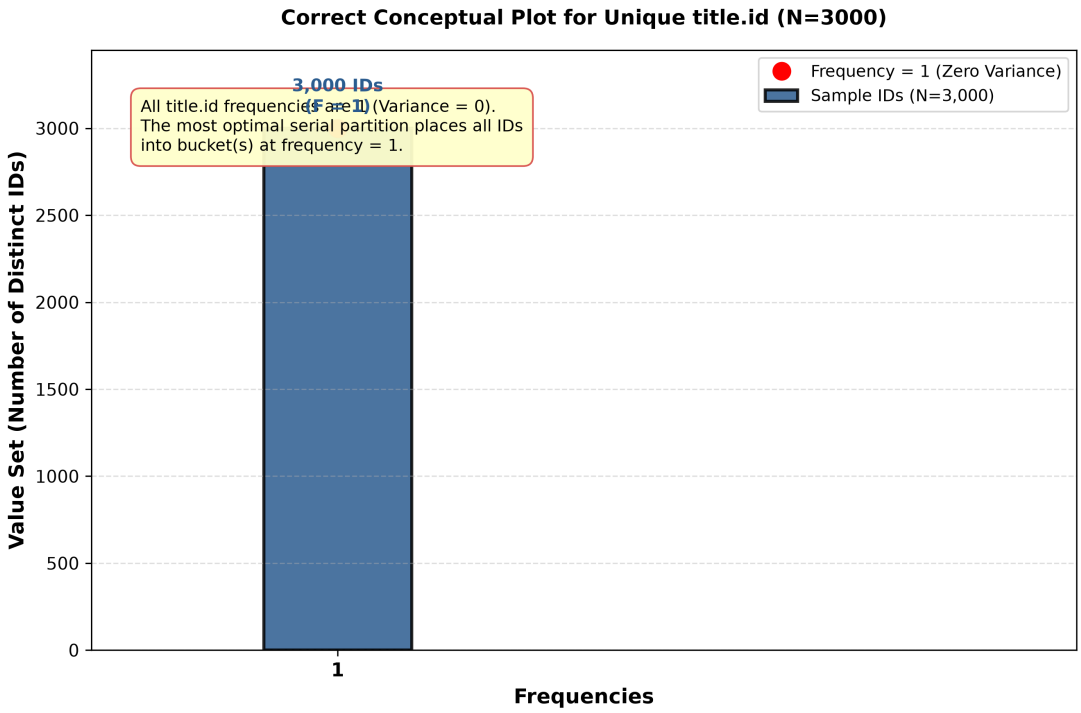


Figure 7: Correct Conceptual Plot for Unique title.id (N=3000)

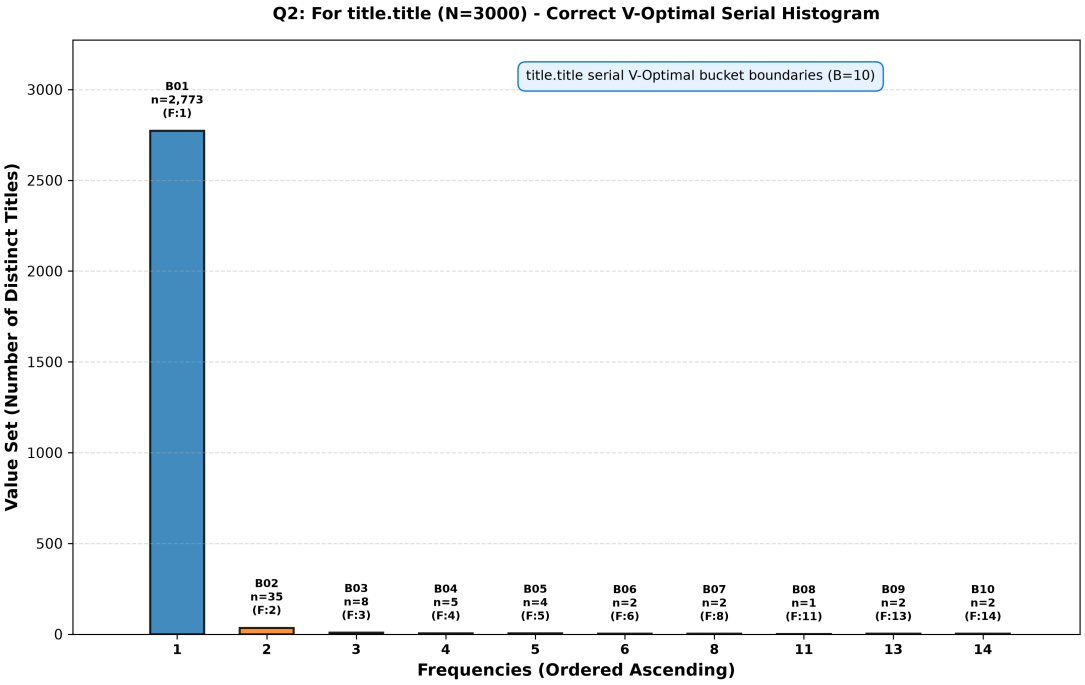


Figure 8: Q2: For title.title (N=3000) - Correct V-Optimal Serial Histogram

B. Sample Size $N = 3000$ Histograms

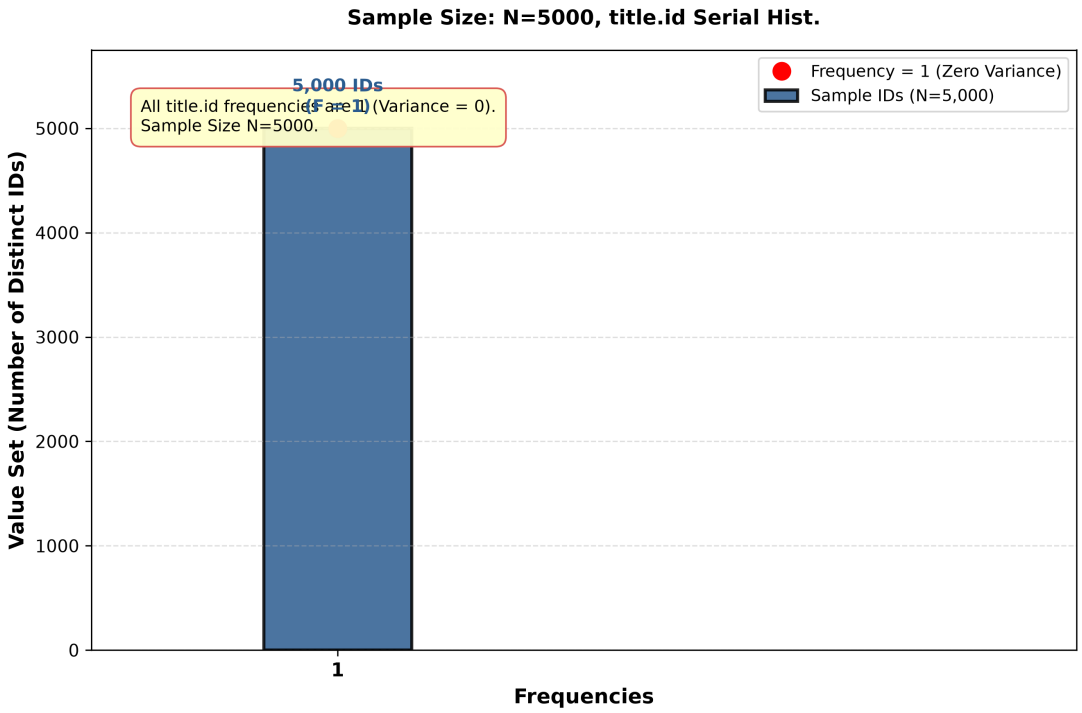


Figure 9: Sample Size: N=5000, title.id Serial Hist.

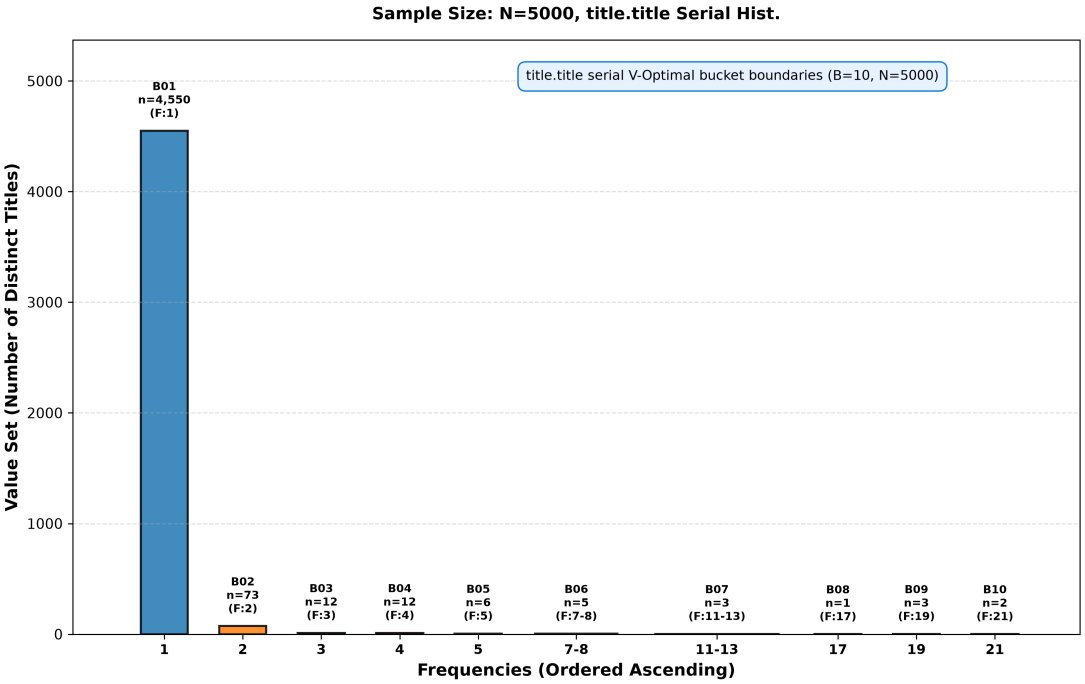


Figure 10: Sample Size: N=5000, title.title Serial Hist.

C. Sample Size $N = 5000$ Histograms

D. Comprehensive 3-Run Summary Table Across Exact Sample Sizes

Exact Sample Size (N)	Column	Run 1 Error	Run 2 Error	Run 3 Error	Average Max Error	Average Build Time
1000	id	0.020667	0.013333	0.016556	0.016852	0.0006s
1000	title	0.637000	0.625000	0.169333	0.477111	0.0015s
3000	id	0.022444	0.009556	0.009889	0.013963	0.0018s
3000	title	0.612556	0.609000	0.616667	0.612741	0.0048s
5000	id	0.013578	0.005778	0.007556	0.008970	0.0029s
5000	title	0.603667	0.600667	0.606400	0.603578	0.0080s

3.2.6 6. Full Step-by-Step Linear Regression Extrapolation ($N = 2,528,312$)

To extrapolate the construction build time for the entire `title` table ($N = 2,528,312$), we fit a linear regression model on the 3-run average measured build times:

$$T(N) = m \cdot N + c$$

A. Sample Data Points (N_i, T_i)

- Point 1: $(N_1, T_1) = (1000, 0.0006)$
- Point 2: $(N_2, T_2) = (3000, 0.0018)$
- Point 3: $(N_3, T_3) = (5000, 0.0029)$

B. Mean Calculations

- $\bar{N} = \frac{1000+3000+5000}{3} = \mathbf{3000}$
- $\bar{T} = \frac{0.0006+0.0018+0.0029}{3} = \mathbf{0.0018}$ seconds

C. Deviations & Sum of Products

N_i	T_i (s)	$(N_i - \bar{N})$	$(T_i - \bar{T})$	$(N_i - \bar{N})^2$	$(N_i - \bar{N})(T_i - \bar{T})$
1000	0.0006	-2000	-0.0012	4,000,000	2.3471
3000	0.0018	0	0.0000	0	0.0000
5000	0.0029	+2000	0.0012	4,000,000	2.3192
Sum				**8,000,000**	**4.6663**

D. Slope (m) and Intercept (c) Formulas

- **Slope (m)**:

$$m = \frac{\sum(N_i - \bar{N})(T_i - \bar{T})}{\sum(N_i - \bar{N})^2} = \frac{4.6663}{8,000,000} = \mathbf{0.00000058} \text{ seconds / sample}$$

- **Intercept (c)**:

$$c = \bar{T} - m \cdot \bar{N} = 0.0018 - (0.00000058 \times 3000) = \mathbf{0.000018} \text{ seconds}$$

E. Extrapolated Build Time for Full Table ($N = 2,528,312$)

$$T(2,528,312) = (0.00000058 \times 2,528,312) + (0.000018) = \mathbf{1.47} \text{ seconds} \quad (\mathbf{0.02} \text{ minutes})$$

3.2.7 7. Sample Size vs Time Relationship Plot

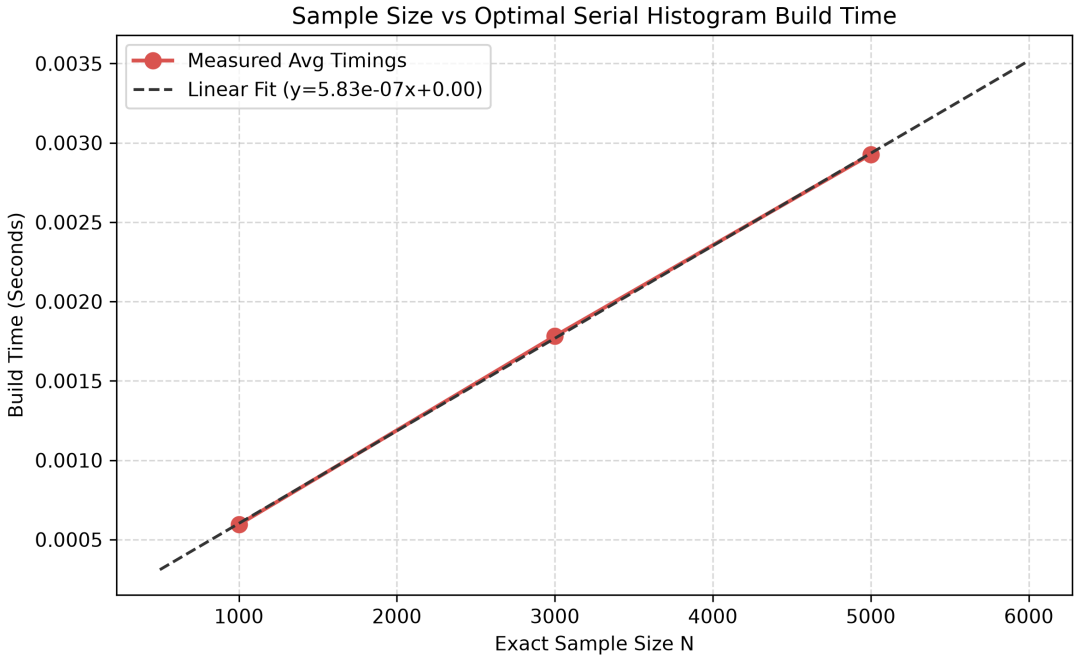


Figure 11: Sample Size vs Time

3.2.8 8. Sample Size vs Max Selectivity Error Relationship Plot

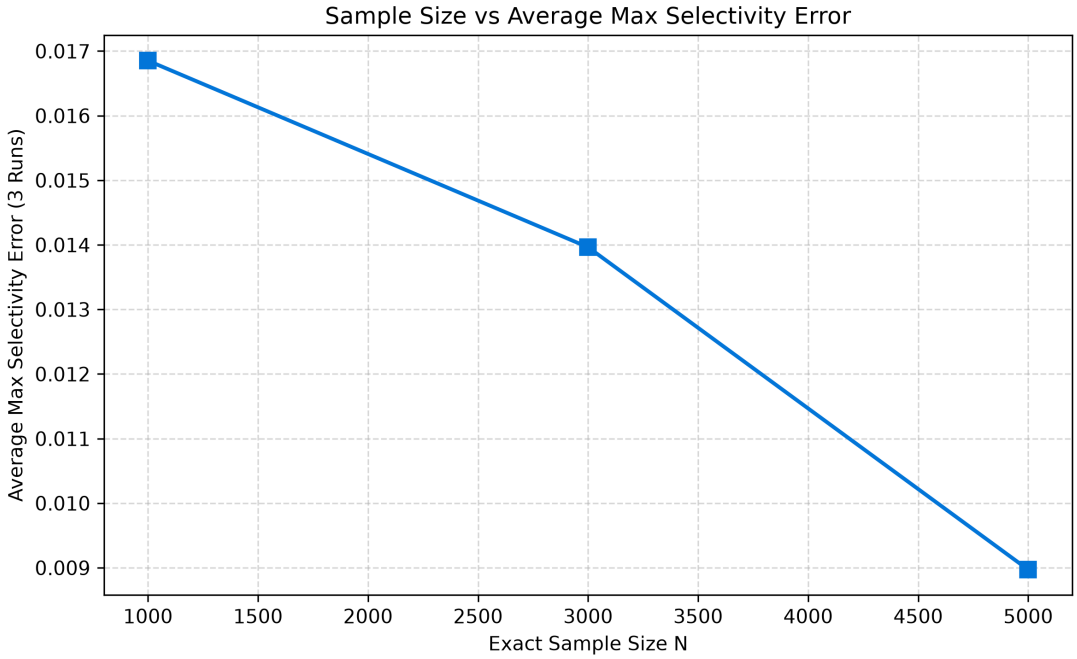


Figure 12: Sample Size vs Max Selectivity Error

3.3 9: The Trade-off of Bucket Counts

Picking the number of buckets B is basically a trade off between how fast the histogram builds and how accurate the query planners estimates are. If we use a small number of buckets, the DP algo runs much faster since its time complexity is O(B*N^2). It also saves memory in the system catalog and keeps query planning times short. The major downside is that wide buckets lead to higher selectivity

errors. The database has to assume the data is uniformly distributed inside those large buckets, which usually isnt true and causes the estimates to lose resolution. On the other hand using more buckets gives the optimizer a highdefinition picture of the actual data distribution, significantly dropping the max selectivity error. The catch is that computing these optimal partitions takes considerably more CPU time, uses more storage space, and adds overhead during the query planning phase before execution can begin.